

The project can be divided into 10 phases.

Overall Workflow

User

- ↳ Enter Company Name / URL
 - ↳ Search (Serper)
 - ↳ Find official website
 - ↳ Extract content
 - ↳ Public Search (Serper)
 - ↳ Merge Information
 - ↳ OpenRouter AI
 - ↳ Generate Summary → Pain Point + Competitors
 - ↳ Display Chart Response
 - ↳ Generate PDF
 - ↳ Send PDF to Discord (optional)

Phase 1 - Project Setup

→ Create the project structure.

Frontend

- React
- Tailwind
- Shadcn/UI

Backend

- Node
- Express

Shared

.env

Readme

Output

Project/
 frontend/
 backend/

(Nothing fancy)

Phase 2 - Build ChatGPT UI

→ This is purely frontend.

Build

- Header
- Chat Window
- Input
- Send Button
- Sidebar (Optional)
- Progress Indicator

(Don't connect APIs yet, just UI)

Phase 3 - Search Engine Integration

→ Now backend start

Input

→ microsoft

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Call

→ serper.dev

Output

→ <https://microsoft.com>

If user already enters url

→ Skip Searching

Output

→ Official Website.

Phase 4 - Website Crawling

- Now crawl url.
- Don't crawl entire internet
- Only discovers pages like

/
about
Products
Services
Contact
pricing
Solutions

- Ignore

login
Signup
privacy
terms
careers

- Output

→ List of important pages.

Phase 5 - Content Extraction

- Visit every page.

- Extract

- Titles
- Paragraphs
- Headings
- Phone
- Address
- Emails
- Products

Note: Clone HTML, Remove

- CSS
- Scripts
- Navigation
- Footer
- Ads

Finally, we have Row Company data.

Phase 6 - Public Research

→ Now use Scrapy ~~again~~ again.

→ Search

- Microsoft headquarters
- Microsoft phone
- Microsoft competitors
- Microsoft products

→ Collect

- Address
- Phone
- Industry
- External information

→ Merge with crawler data.

Now we have, structured company data.

Phase 7 - AI Analysis

→ Now send everything to open Router.

→ Prompt:

Here is company information.

Generate:

- Summary
- Products
- Pain points
- Competitors
- Industry

→ Output JSON Only.

→ AI Return

- Summary
- Pain points
- Competitors.

Phase 8 - Show Result

~~to~~ → Now frontend display.

→ Company card

→ Summary

→ Products

→ Pain Points

→ Competitors

→ Sources

(Like ChatGPT Response.)

Phase 9 - PDF Generation

→ Now generate professional PDF

→ Include

→ Logo

→ Company

→ Website

→ Address

→ Phone

→ Summary

→ Products

→ Pain Points

→ Competitors

→ Button

→ Download PDF

Phase 10 - Discord (Bonus)

- Settings page
 - Bot Token
 - Channel ID
- After PDF Generated, automatically send
 - Application Name
 - Applicant Email
 - Company Name
 - Website
 - PDF

Complete Data Flow:

